

ERDENEMANDAL, Mongolia

WMO: 442370

Lat: 48.5294N

Lon: 101.3675E

Elev: 1509

StdP: 84.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-29.5	-34.7	0.2	-30.6	-32.7	0.2	-28.3	10.5	-12.2	9.3	-12.2	1.8	230	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	27.7	16.9	25.6	15.9	23.9	15.1	18.5	25.8	17.1	23.8	16.0	22.2	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	13.5	23.0	14.5	12.4	20.7	13.4	11.6	19.5	59.4	25.8	54.2	23.9	50.7	22.2	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.3	9.2	8.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-33.2	31.2	3.2	2.2	-35.5	32.8	-37.3	34.1	-39.1	35.3	-41.4	36.9
			-33.2	20.3	3.1	2.0	-35.5	21.8	-37.4	22.9	-39.1	24.1	-41.4	25.5
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-18.4	-14.0	-5.8	3.1	9.1	14.6	16.9	14.8	9.0	1.0	-8.1	-15.0	(6)
(7)		DBStd	13.12	6.04	6.17	6.51	5.16	4.75	3.58	2.87	3.09	4.50	5.28	6.54	5.99	(7)
(8)		HDD10.0	3998	880	672	488	211	74	6	0	3	68	279	543	774	(8)
(9)		HDD18.3	6475	1138	905	747	457	287	121	63	116	279	537	793	1032	(9)
(10)		CDD10.0	597	0	0	0	4	46	143	213	151	39	1	0	0	(10)
(11)		CDD18.3	32	0	0	0	0	1	7	17	6	1	0	0	0	(11)
(12)		CDH23.3	505	0	0	0	0	33	147	218	94	12	0	0	0	(12)
(13)		CDH26.7	106	0	0	0	0	4	29	52	20	1	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.3	2.5	3.2	3.7	3.7	2.8	2.3	2.4	2.7	2.9	2.7	2.5	(14)
(15)	Precipitation	PrecAvg	282	2	2	2	9	20	56	80	70	22	8	3	2	(15)
(16)		PrecMax	397	8	7	11	34	59	117	204	184	51	39	13	15	(16)
(17)		PrecMin	177	0	0	0	0	1	17	18	11	2	0	0	0	(17)
(18)		PrecStd	65	2	2	2	9	14	28	35	37	12	8	3	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	4.1	13.3	20.6	26.8	29.8	31.1	29.1	25.0	17.2	9.5	1.0	(19)
(20)		MCWB	-4.2	-0.2	5.4	9.5	13.6	17.9	19.5	19.0	13.3	8.9	3.6	-2.5	-2.5	(20)
(21)		2%	DB	-4.5	0.8	9.7	17.2	23.7	27.0	28.0	25.9	21.9	14.4	6.1	-2.3	(21)
(22)		MCWB	-6.8	-2.8	3.3	8.3	12.4	16.5	17.7	16.7	12.9	7.5	1.5	-5.1	-5.1	(22)
(23)		5%	DB	-7.1	-2.0	6.8	14.6	21.2	24.8	25.6	23.6	19.5	11.9	3.3	-4.5	(23)
(24)		MCWB	-9.0	-4.8	1.4	6.7	11.3	15.1	16.7	15.7	11.9	5.7	-0.4	-6.8	-6.8	(24)
(25)		10%	DB	-9.9	-4.7	4.0	12.3	18.7	22.9	23.7	21.6	17.2	9.4	0.8	-6.6	(25)
(26)		MCWB	-11.4	-7.0	-0.5	5.4	9.8	14.0	15.9	14.8	10.6	4.3	-2.2	-8.5	-8.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.8	0.1	6.3	11.2	15.9	19.2	21.4	19.9	15.4	9.7	3.8	-2.3	(27)
(28)		MCDWB	-1.3	3.8	12.3	18.9	24.7	27.7	29.8	28.3	22.8	16.2	8.8	0.8	0.8	(28)
(29)		2%	WB	-6.7	-2.7	3.6	8.8	13.8	17.2	19.1	17.5	13.6	7.8	1.6	-5.0	(29)
(30)		MCDWB	-4.5	0.6	9.0	15.8	21.8	25.1	26.0	24.5	20.4	13.8	5.7	-2.3	-2.3	(30)
(31)		5%	WB	-8.9	-4.8	1.5	7.2	11.8	15.9	17.6	16.2	12.3	6.0	-0.3	-6.8	(31)
(32)		MCDWB	-7.1	-2.1	6.5	14.1	19.8	23.5	24.0	22.2	18.7	11.4	3.1	-4.6	-4.6	(32)
(33)		10%	WB	-11.3	-7.0	-0.3	5.6	10.2	14.7	16.4	15.2	10.8	4.3	-2.2	-8.5	(33)
(34)		MCDWB	-9.9	-4.8	3.9	11.8	17.5	21.5	22.2	20.4	16.9	9.1	0.7	-6.6	-6.6	(34)
(35)	Mean Daily Temperature Range		MDBR	10.2	12.6	13.2	12.5	13.2	12.8	11.4	11.6	12.8	12.1	10.7	9.8	(35)
(36)		5% DB	MCDBR	12.3	15.1	15.4	15.9	17.2	16.8	15.5	15.4	16.0	15.5	13.0	11.3	(36)
(37)			MCWBR	10.6	12.3	10.7	9.8	9.5	8.6	7.5	8.1	9.1	10.1	9.6	9.2	(37)
(38)			MCDBR	12.3	14.9	15.1	15.4	16.1	15.2	13.9	13.7	15.2	15.1	12.7	11.1	(38)
(39)		5% WB	MCWBR	10.6	12.2	10.9	10.3	10.1	8.8	7.6	7.8	9.2	10.3	9.5	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.232	0.256	0.300	0.341	0.357	0.361	0.351	0.332	0.297	0.276	0.254	0.234	(40)	
(41)		taud	2.368	2.330	2.276	2.260	2.259	2.351	2.430	2.455	2.492	2.468	2.390	2.340	(41)	
(42)		Ebn at Noon	867	914	915	907	903	899	903	909	915	881	824	808	(42)	
(43)		Edh at Noon	65	87	109	124	130	119	109	101	88	75	64	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.77	2.86	4.37	5.58	6.42	6.42	6.11	5.35	4.56	3.06	1.88	1.42	(44)	
(45)		RadStd	0.07	0.09	0.15	0.26	0.38	0.31	0.36	0.22	0.19	0.16	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	-2.02	N/A	-0.60	-0.74	N/A	N/A	N/A	N/A

Nomenclature: See separate page